

班级 2203254w

学号 22009201185

本科毕业设计（论文）
外文资料翻译

毕业设计题目	智能客服机器人的设计
	与交互优化
外文资料题目	Artificial Intelligence and Machine Learning in
	Financial Services (节选)
学院	计算机科学与技术学院
专业	软件工程
学生姓名	王奇
指导教师姓名	谢海明，魏昱恒

第三篇 外文资料

Artificial Intelligence and Machine Learning in Financial Services (节选)

资料来源：Paul Tierno. Artificial Intelligence and Machine Learning in Financial Services.

CRS Report R47997, Congressional Research Service, April 3, 2024. 本文节选 Introduction、Technology Overview in Brief、Finance and Artificial Intelligence、AI/ML Applications in Finance 等部分，译文仅用于课程作业，非官方译本。

原文

Artificial Intelligence and Machine Learning in Financial Services

The financial industry's adoption of artificial intelligence (AI) and machine learning (ML) is evolving as financial firms employ ever greater levels of technology and automation to deliver services. Expanding on earlier models of quantitative analysis, AI/ML has often been adopted in finance to solve discrete challenges, such as maximizing profit and minimizing risk. Yet the industry's adoption of the newer technology also occurs against perceptions that are steeped in tradition and historical financial regulation, and regulators want to ensure that the technology does not sidestep regulations frequently described as technology neutral.

Technological advances in computer hardware, capacity, and data storage—which permit the collection and analysis of data—helped fuel the development and use of AI/ML technologies in finance. Unlike older algorithms that automated human-coded rules, new AI models can learn by themselves and make inferences and recommendations not identified by modelers in advance. This shift in technology has also enabled the use of new types of data including alternative data, unstructured data, and unlabeled information data—which, when combined, extend the technologies' uses to new financial services or products.

Different parts of the financial services industry have adopted AI/ML technology to varying degrees and for various purposes. Some uses of AI/ML include powering chatbots in customer service functions, identifying investment opportunities and/or executing trades, augmenting lending models or making lending decisions, and identifying and preventing fraud. The extent to which a sector or firm adopts various technologies reflects a variety of factors, including a firm's ability to fund internal development and regulatory requirements.

Financial industry policymakers face competing pressures. Financial service providers and technology companies are likely to continue adopting and promoting AI/ML to save time and money and promote accessibility, accuracy, and regulatory compliance. However, challenges and risks in the form of bias, potential for systemic risk and manipulation, affordability, and consequences for employment remain. Determining whether the existing regulatory structure is sufficient—or whether one that is more closely tailored to the technological capacities of the evolving technology is necessary—has emerged as a key consideration. Should Congress consider the legislative framework governing AI/ML in finance, industry and consumers alike will expect that it weighs the benefits of innovation with existing and potential future challenges and risks.

Introduction

Artificial intelligence (AI) is the general term used to describe the process of programming computers and machines to think and operate like humans. Machine learning (ML) is a subset of AI that describes computers and programs that may be programmed to operate with minimal human intervention and can in some instances learn and/or update themselves. Various events over the past few years have helped raise the profile of AI/ML and its role in delivering financial services. OpenAI's introduction of the large language model (LLM) ChatGPT in 2022 was a rare moment when an AI/ML technology became directly accessible by the broad public. While momentum around AI/ML had been building over the past decade at least, the ubiquity of competing prominent technologies concentrated attention on all manner of applications, including those in the financial sector.

AI/ML is accelerating a long, ongoing shift in finance from face-to-face interactions conducted in a customer's community to online products and services fueled by advanced algorithms that require little or no human interaction and can occur anywhere. Such applications may have benefits for financial institutions and their clients. Relative to human operation or models built using earlier technology, AI/ML provision of financial services may be faster and cheaper due to their speed and efficiency, and they may be able to expand service to more individuals due to their ability to analyze alternative data and find latent connections. The financial industry's use of technology is not new, and there are elements to its adoption of AI/ML that resemble past periods.

Adoption of the technology also brings challenges and risks. While AI may eliminate some human bias in financial services, it may also introduce or exacerbate bias. Certain AI-market specific factors—such as using the same data to train models and concentration of technology and technological capacity—may encourage monocultures, or herd-like behavior, that introduce systemic risk. The ability of certain models to learn on their own, free from human oversight, may also allow them to engage in market manipulation, conflicts of interest, or other unlawful activity.

This report discusses the role of AI/ML in financial services broadly. First, it provides a brief overview of some technical terms. Next, it discusses the motivations for the financial industry's use of AI/ML. It then provides an overview of some financial applications of the technology. Finally, it addresses the regulatory framework and considers various policy issues, including any applicable regulations.

Technology Overview in Brief

AI is the broad label for technologies that give computer systems the ability to learn new concepts or tasks and to reason and draw useful conclusions about the world. Earlier versions of AI were largely dependent on their human developers that programmed machines with a series of deterministic rules, such as if-then statements, alongside troves of subject matter content. Such technology required specific syntax for questions or commands and produced responses on topics for which it had been programmed. When faced with a situation for which it had not been programmed, it would freeze. That dependency would likely disqualify it from the AI in current taxonomy and is largely disappearing. According to some, most applications of AI now are ML, so the terms as used are synonymous.

ML is a subset of AI focused on constructing computer systems that can automatically improve through experiencing and determining certain characteristics inherent in all learning systems, which can include computers, humans, and organizations. ML attempts to give computers the ability to learn and change without being reprogrammed. It uses algorithms—sequenced actions designed to solve specific problems—that improve automatically with various degrees of, and often little or no, human interaction. In other words, unlike earlier AI, it is meant to be adaptable. ML systems are intended to determine relationships among variables or recognize patterns in large data sets. Yet ML is not one thing, such as a discrete computer application, nor is it even

one technology. Instead, it is a group of systems that is defined in part by the type of information used to train the model and the amount of involvement provided by human trainers.

Technological developments in computing and telecommunications have been a boon to AI/ML, and the financial industry's use of it includes advanced performance computing, which includes hardware, cloud and edge computing, and communication technologies. However, the financial industry's adoption of AI/ML depends upon the availability of data, including alternative data not traditionally used in financial decisionmaking, and the ability to process and analyze unstructured data. These technological advances move in tandem with the amassing of large quantities of data that require scalable architecture for efficient storage, manipulation, and analysis.

Some of today's technologies have their conceptual origins decades ago, and they are now being developed. Unsupervised learning's ability to process unlabeled data may identify relationships—for example, in market interactions or among loan applications—where its programmers had not thought to look. Similarly, the ability to process unstructured and alternative data that previously could not be analyzed—such as images and social media posts—has opened it to analysis.

Big Data

Big data refers both to a type of information and how it is used. It refers to large data sets that exhibit specific characteristics such as volume, velocity, variety, and/or variability and requires scalable architecture to store, manage, and analyze. The term also generally refers to the data use by managers, companies, and industries to facilitate business decisions.

Some advances in ML have been directly attributed to the ability to process more data. The specific big data used in finance depends on the sector or use case. For example, some firms may use big data—including social media—when deciding whether to make loans. In a capital markets and trading environment, big data may include continuous data feeds from various stock exchanges, which can include stock prices and broader market moves and histories of transactions. Data in this environment may also include company-level data that may influence a stock performance, including financial statements and quarterly and annual regulatory filings. Alternative data refers to information that has not traditionally been used to make financial decisions. In consumer credit underwriting, alternative data refers to non-credit data that has not traditionally been used in determining credit scores, such as rent and utility payments. Banks and

credit card and payment companies may also use alternative and unstructured data—such as transaction information and location or GPS data—in their fraud detection and anti-money laundering activities. Asset managers and other capital markets participants may also use GPS and satellite imagery of retailer parking lots or social media mentions that provide insight into a company's performance. The availability of different forms of data and the models they use have expanded the ways financial firms may use both.

Recent interest in AI has been marked by a period of increased investment in and development and adoption of AI/ML technologies by various sectors, including the financial industry.

According to one bank's research, investment in AI is expected to climb to around \$100 billion in the United States in 2025 and nearly \$200 billion globally. Notably, there is likely broad investment from traditional financial institutions, with finance-focused startups attracting investment from venture capital firms.

Finance and Artificial Intelligence

According to the Organisation for Economic Co-operation and Development (OECD), AI is powering digital transformation that has been rapidly deployed in finance. Adoption of this newer technology is not surprising as finance has often been linked with cutting-edge technology. However, the industry is also known for its use of often obsolete technology and associations with traditional practices that are at odds with new technologies. While in the past, personal banking was rooted in community relationships and conducted in person, now nearly all transactions can be performed online. In capital markets transactions, electronic trading has essentially ended the need for trading to be conducted person to person at the physical location of exchanges. Meanwhile, the use of AI/ML, like the related adoption of a broad set of financial technology, also has implications for regulation, which prioritizes industry transparency, explainability, and fairness. Finance's adoption of AI/ML thus takes place as technology and its capacity may be evolving more rapidly than societal and regulatory expectations.

Motivation behind the financial sector's adoption of AI is not new. Finance's adoption of AI has been compared to the development of net-present-value calculations, the invention of the automated teller machine, use of expert systems for personal financial planning, and automated trading, all of which employed different technologies.

In addition to a general gravitation to technology, various characteristics of the financial industry have made it a ripe testing ground for AI/ML. Information processing is the central function of

financial markets. For example, AI/ML generally requires large quantities of data on which to train its systems. Data is a quintessential feature across the various financial sectors. The industry amasses data from economic indicators, financial markets, individual consumers and businesses, and payments and transactions, among countless other sources. Moreover, while comprised of different subsectors—with diverse functions, services, and products—finance presents quantifiable problems, the solutions for which are often reduced to maximizing returns considering a given risk tolerance while considering some number of other quantifiable variables. As such, the industry is generating large amounts of data and providing the operating environment for AI/ML technologies.

AI/ML may also increase the speed with which financial transactions may take place. For example, high-frequency trading, which is also increasingly automated and programmatic, allows trades to occur in the smallest fractions of a second, speeds at which neither human analysis nor execution are possible. Also, if a financial institution uses AI/ML to determine whether it should make a loan, the speed may be evident in multiple ways. AI/ML-induced speed may allow financial institutions to update their lending models. Lenders may also use AI/ML-based lending models, integrated with digital platforms and web interfaces, to deliver decisions to customers more quickly.

AI/ML Applications in Finance

Financial industry and market participants use AI for a variety of functions. These functions have multiple purposes, including improving efficiency, assisting with decisionmaking, producing analysis and forecasting, increasing profitability, managing risk, and underwriting credit. Moreover, they may be used in support functions or front office functions. Back office applications may include post-trade processing, trading profit and loss reconciliations, and data analytics. Other applications may include regulatory compliance functions, such as know-your-customer checks and customer identification programs, anti-money laundering and countering the financing of terrorism programs, and anti-fraud. AI may also assist in asset allocation, robo-advising, and trade execution. Many of the actions taken by financial firms are multistep and closely related, and AI may be used for and benefit from different parts of a business or phases of a business process.

Credit underwriting—assessing the risk of prospective borrowers defaulting on loan repayment—is one of banks' and other financial institutions' primary businesses. Loan-making financial

institutions have automated the methods for making loan decisions, including through the use of electronic data. For consumer loans, such methods generally entail using a mathematical formula called a scoring model to determine a consumer's credit score and subsequently whether the firm should make the loan and at what interest rate. Various factors—including bill paying history, unpaid debt, outstanding loans, and accounts—go into determining a credit score. Recently, however, some financial companies have considered using AI/ML to augment or replace traditional credit scoring.

Lenders have been using predictive models for decades. Traditionally, statistical regression methods used data from the credit reporting bureaus, assigning weights to different variables to help forecast whether an applicant would default on a loan and to determine the likelihood of on-time loan repayment. More recently, firms may use ML-based measures based on their ability to analyze large amounts and different types of data—including transaction data—and their ability to discern other important relationships not visible in traditional models. Adoption of AI/ML for credit underwriting may differ among banks and nonbank fintech. Various regulatory requirements may make banks hesitant to use the technology for credit underwriting decisions. Generally, because nonbanks may have different regulatory requirements and supervision than banks do, they may be more willing to use AI/ML technology.

Chatbots are computer programs that interact with people online by simulating human conversation through text and voice commands. Banking chatbots provide immediate assistance 24/7, reducing wait times, addressing customer inquiries, providing information on account balances and transaction history, and guiding users through various banking processes. Chatbots used in banking provide substantial cost savings annually, according to one estimate.

Regtech, or regulatory technology, refers to regulated financial institutions' use of technology to meet various regulatory, compliance, and data reporting functions. Regtech has existed for around a decade. It had focused on compliance with regulations related to onboarding and identifying new customers, but more recently it has been described as being instrumental in anti-money laundering, countering the financing of terrorism, fraud prevention, risk management, stress testing, and micro and macroprudential reporting. For example, banks and other financial institutions use robotics process automation to help them comply with reporting requirements and repopulating required data on a regular basis. In many cases, banks' and other financial institutions' adoption of regtech may involve partnering with third-party providers with expertise in the field.

A key form of regtech aided by AI/ML is detecting, preventing, and reporting unauthorized and illicit financial activities for banks and other financial institutions. The shift to AI/ML solutions is due to its more adaptable approaches and its ability to leverage more types of data. Banks and other financial institutions must detect, prevent, and report on unauthorized and illicit financial activity, and there has been an increase in the number of financial institutions using AI/ML to address the issue. Banks and other financial institutions can train models on huge volumes of consumer behavior data they generate, allowing the ML models to learn fraud patterns and to then detect fraudulent behavior in practice. Similarly, there may be benefits to using the technology across the anti-money laundering value chain, including at onboarding and client screening and with particular and immediate benefits coming from transaction monitoring. In addition, researchers suggest that the technology is also useful in reducing false positives, freeing banks to dedicate resources to actual instances of fraud.

Natural language processing (NLP) is a branch of AI that trains computers to understand and process language as humans do. NLP can be used to develop LLMs that can create content in the form of essays or computer code from prompts using techniques that predict the next word in a sequence of words based on certain probabilities. This technology supports applications in finance, including powering chatbots, surveying companies' own large troves of data, and writing code. LLMs are examples of artificial neural networks that attempt to reflect the functioning of the human brain by replicating neurons as mathematical representations and organizing layers of such artificial neurons into clusters or layers. Aside from finance-specific applications, LLMs and ChatGPT in particular provided general public audiences with some of the first concrete and directly applicable examples of AI that until now had been reserved for specialists.

中文

金融服务中的人工智能与机器学习（节选）

资料节选范围：Introduction；Technology Overview in Brief；Finance and Artificial Intelligence；AI/ML Applications in Finance。

金融服务中的人工智能与机器学习

金融行业对人工智能（AI）和机器学习（ML）的采用正在不断发展。金融机构在提供服务时使用越来越高水平的技术和自动化工具。AI/ML 延续了早期定量分析模型的思路，常被用于解决相对明确的金融问题，例如最大化收益和最小化风险。不过，金融行业采用新技术的过程，也处在传统观念和历史金融监管框架之中。监管者希望确保这些技术不会绕开通常被描述为技术中立的监管要求。

计算机硬件、计算能力和数据存储方面的技术进步，使数据收集和分析成为可能，也推动了 AI/ML 技术在金融领域的发展。与早期自动执行人工编码规则的算法不同，新的 AI 模型能够自行学习，并作出建模人员事先没有明确设定的推断和建议。这种技术变化也使金融机构能够使用新的数据类型，包括替代数据、非结构化数据和无标签信息数据。它们结合在一起，可以把相关技术扩展到新的金融服务或产品之中。

金融服务行业的不同领域，以不同程度、出于不同目的采用 AI/ML 技术。AI/ML 的应用包括驱动客户服务中的聊天机器人，识别投资机会或执行交易，增强信贷模型或作出贷款决策，以及识别和预防欺诈。某一行业或企业采用不同技术的程度，受到多种因素影响，其中包括企业自身开展内部研发的资金能力以及监管要求。

金融行业政策制定者面临相互竞争的压力。金融服务提供者和技术公司很可能继续采用和推广 AI/ML，以节省时间和成本，并提升可获得性、准确性和合规能力。然而，偏见、系统性风险和市场操纵可能性、可负担性以及就业影响等挑战和风险仍然存在。现

有监管结构是否足够，或者是否需要建立更贴合新技术能力的监管框架，已经成为重要问题。如果立法机关考虑金融领域 AI/ML 的制度框架，行业和消费者都会期待其在创新收益与既有及潜在风险之间取得平衡。

引言

人工智能是一个总称，用于描述把计算机和机器编程为像人一样思考和运行的过程。机器学习是人工智能的一个子集，指计算机和程序可以在最少人工干预下运行，并且在某些情况下能够学习或自我更新。过去几年发生的多个事件，提高了 AI/ML 及其在金融服务交付中作用的关注度。OpenAI 于 2022 年推出大型语言模型 ChatGPT，是 AI/ML 技术直接触达广大公众的少见时刻。虽然围绕 AI/ML 的发展势头至少已经持续了十多年，但多个重要技术的普及，使人们把注意力集中到各类应用之上，其中也包括金融领域的应用。

AI/ML 正在加速金融业长期持续的转变，即从客户所在社区中的面对面互动，转向由高级算法驱动的线上产品和服务。这些产品和服务几乎不需要人工互动，并且可以在任何地点发生。相关应用可能给金融机构及其客户带来收益。与人工操作或使用早期技术构建的模型相比，AI/ML 提供金融服务可能因为速度和效率而更快、更便宜；由于能够分析替代数据并发现潜在联系，它们还可能把服务扩展到更多个人。金融行业使用技术并不是新现象，其采用 AI/ML 的某些方面也与以往技术周期相似。

采用这类技术同样带来挑战和风险。AI 可能消除金融服务中的某些人为偏见，但也可能引入或加剧偏见。某些 AI 市场特定因素，例如使用相同数据训练模型，以及技术和技术能力的集中，可能鼓励单一化或类似羊群效应的行为，从而引入系统性风险。某些模型可以在缺乏人工监督的情况下自行学习，这也可能使其从事市场操纵、利益冲突或其他违法活动。

本报告广泛讨论 AI/ML 在金融服务中的作用。首先，它简要介绍若干技术术语；其次，讨论金融行业使用 AI/ML 的动机；随后，概述该技术在金融中的若干应用；最后，报告讨论监管框架，并考虑包括适用监管在内的多个政策问题。

技术概览

AI 是一个宽泛标签，指赋予计算机系统学习新概念或任务、进行推理并对世界得出有用结论能力的技术。早期 AI 在很大程度上依赖开发者，开发者需要用一系列确定性规则，例如 if-then 语句，并结合大量专业内容来编写机器。此类技术要求问题或命令采用特定语法，并只能对已经被编程的主题作出回应。当遇到没有被编程的情境时，它就会停滞。这种依赖性在当前分类中很可能不足以被认定为 AI，并且正在逐渐消失。有观点认为，现在大多数 AI 应用都是 ML，因此二者在实际使用中往往具有相近含义。

ML 是 AI 的一个子集，重点在于构建能够通过经验自动改进的计算机系统，并识别所有学习系统中固有的某些特征，这些学习系统可以包括计算机、人类和组织。ML 试图赋予计算机在无需重新编程的情况下学习和变化的能力。它使用算法，即为解决特定问题而设计的一系列步骤，并在不同程度、通常是很少甚至没有人工互动的情况下自动改进。换言之，与早期 AI 不同，ML 具有适应性。ML 系统旨在确定变量之间的关系，或识别大型数据集中的模式。不过，ML 并不是某一个离散的计算机应用，甚至也不是一种单一技术。它是一组系统，部分由训练模型所使用的信息类型以及人工训练者参与程度来定义。

计算和通信技术的发展极大促进了 AI/ML，金融行业对相关技术的使用也包括高性能计算，其中涵盖硬件、云计算、边缘计算和通信技术。不过，金融行业采用 AI/ML 还取决于数据可得性，包括传统金融决策中不使用的替代数据，以及处理和分析非结构化数

据的能力。这些技术进步与海量数据积累同步发生，而海量数据需要可扩展架构来实现高效存储、处理和分析。

今天某些技术的概念来源可以追溯到几十年前，而它们如今正在得到进一步发展。无监督学习处理无标签数据的能力，可以识别程序员原本没有想到去寻找的关系，例如市场互动之间的关系或贷款申请之间的关系。同样，对以往无法分析的非结构化数据和替代数据进行处理的能力，例如图像和社交媒体帖子，也使其能够进入分析过程。

大数据

大数据既指一种信息，也指这种信息的使用方式。它指具有容量、速度、多样性或可变性等特征的大型数据集，并需要可扩展架构来存储、管理和分析。该术语通常也指管理者、公司和行业使用数据来辅助商业决策。

ML 的一些进步直接归因于处理更多数据的能力。金融中使用的具体大数据取决于行业或使用场景。例如，一些公司在决定是否发放贷款时，可能使用包括社交媒体在内的大数据。在资本市场和交易环境中，大数据可能包括来自不同证券交易所的连续数据流，其中包含股票价格、更广泛的市场波动以及交易历史。这一环境中的数据还可能包括影响股票表现的公司层面数据，例如财务报表以及季度和年度监管文件。

替代数据指以往没有被用于金融决策的信息。在消费者信贷承销中，替代数据是指以往没有用于确定信用评分的非信贷数据，例如租金和水电费支付记录。银行、信用卡公司和支付公司还可能在欺诈检测和反洗钱活动中使用替代数据和非结构化数据，例如交易信息、位置数据或 GPS 数据。资产管理者和其他资本市场参与者也可能使用零售商停车场的 GPS 和卫星图像，或社交媒体提及信息，以洞察公司表现。不同形式的数据及其所使用的模型的可得性，扩展了金融企业使用二者的方式。

近期对 AI 的关注，伴随着各行业对 AI/ML 技术投资、开发和采用的增加，金融行业也包括在内。根据某银行研究，美国 AI 投资预计将在 2025 年升至约 1000 亿美元，全球投资接近 2000 亿美元。值得注意的是，传统金融机构很可能进行广泛投资，而面向金融的初创企业也吸引了风险投资机构的资金。

金融与人工智能

根据经济合作与发展组织的说法，AI 正在推动数字化转型，并已在金融领域快速部署。金融业采用这种新技术并不令人意外，因为金融行业常常与前沿技术相联系。然而，该行业也以使用陈旧技术，以及与新技术相冲突的传统实践相联系。过去，个人银行业务植根于社区关系，并通过面对面方式完成；如今，几乎所有交易都可以在线完成。在资本市场交易中，电子交易基本消除了必须在交易所现场通过人与人方式进行交易的需要。与此同时，AI/ML 的使用，类似于更广泛金融科技的采用，也会影响监管，因为监管强调行业透明性、可解释性和公平性。因此，金融领域采用 AI/ML 的过程，发生在技术及其能力发展可能快于社会和监管预期的背景下。

金融部门采用 AI 的动机并不新鲜。金融业采用 AI 曾被拿来与净现值计算的发展、自动柜员机的发明、用于个人理财规划的专家系统以及自动化交易进行比较，这些都曾经使用不同技术。

除了总体上倾向于使用技术之外，金融行业的若干特征也使其成为 AI/ML 的适宜试验场。信息处理是金融市场的核心功能。例如，AI/ML 通常需要大量数据来训练系统。数据是各金融部门的典型特征。该行业从经济指标、金融市场、个人消费者和企业、支付与交易等无数来源积累数据。此外，金融行业由不同子行业构成，具有多样功能、服务和产品，但金融问题通常是可量化的，其解决方案常被简化为在既定风险容忍度和其

他若干可量化变量约束下最大化收益。因此，金融行业既生成大量数据，也为 AI/ML 技术提供运行环境。

AI/ML 还可能提高金融交易发生的速度。例如，高频交易也越来越自动化和程序化，它允许交易在极短时间内发生，其速度是人类分析或执行都无法达到的。同样，如果金融机构使用 AI/ML 来判断是否发放贷款，这种速度也可能以多种方式体现出来。AI/ML 带来的速度可能使金融机构更快更新其贷款模型。贷款机构还可以把基于 AI/ML 的贷款模型与数字平台和网络界面集成，以更快向客户提供决策。

AI/ML 在金融中的应用

金融行业和市场参与者把 AI 用于多种功能。这些功能具有多重目的，包括提高效率、辅助决策、产生分析和预测、增加盈利能力、管理风险以及进行信贷承销。它们既可以用于支持性职能，也可以用于前台职能。后台应用可能包括交易后处理、交易盈亏核对和数据分析。其他应用可能包括监管合规职能，例如了解你的客户检查、客户身份识别项目、反洗钱和反恐融资项目以及反欺诈。AI 还可以辅助资产配置、智能投顾和交易执行。金融企业采取的许多行动都是多步骤且紧密相关的，AI 可以用于业务的不同部分或流程阶段，并从中受益。

信贷承销，即评估潜在借款人违约风险，是银行和其他金融机构的主要业务之一。发放贷款的金融机构已经通过电子数据等方式，使贷款决策方法自动化。对于消费者贷款，这些方法通常需要使用一个称为评分模型的数学公式来确定消费者信用评分，并进一步决定企业是否发放贷款以及采用何种利率。账单支付历史、未偿债务、未结清贷款和账户等多种因素都会影响信用评分。不过，近来一些金融公司开始考虑使用 AI/ML 来增强或替代传统信用评分。

贷款机构已经使用预测模型几十年。传统上，统计回归方法使用来自信用报告机构的数据，为不同变量分配权重，以帮助预测申请人是否会贷款违约，并确定按时还款的可能性。最近，企业可能使用基于 ML 的度量方法，因为这类方法能够分析大量且不同类型的数据，包括交易数据，并识别传统模型中不可见的其他重要关系。不同机构采用 AI/ML 进行信贷承销的情况可能不同。各种监管要求可能使银行在使用该技术作出信贷承销决策时更加谨慎。一般而言，由于非银行金融科技机构面临的监管要求和监管方式可能不同于银行，它们可能更愿意使用 AI/ML 技术。

聊天机器人是与人在线互动的计算机程序，可以通过文本和语音命令模拟人与人的对话。银行聊天机器人能够全天候提供即时帮助，减少等待时间，回应客户咨询，提供账户余额和交易历史信息，并引导用户完成各种银行流程。根据一项估计，银行中使用的聊天机器人每年能够带来可观的成本节约。

监管科技是指受监管金融机构使用技术来满足各种监管、合规和数据报告职能。监管科技已经存在约十年。它最初侧重于与新客户 onboarding 和客户识别有关的监管合规，近来则被认为在反洗钱、反恐融资、欺诈预防、风险管理、压力测试以及微观和宏观审慎报告方面发挥重要作用。例如，银行和其他金融机构使用机器人流程自动化，帮助其遵守报告要求，并定期重新填充所需数据。在许多情况下，银行和其他金融机构采用监管科技，可能需要与拥有该领域专业能力的第三方提供商合作。

由 AI/ML 支持的一种重要监管科技形式，是为银行和其他金融机构检测、预防并报告未经授权和非法金融活动。转向 AI/ML 解决方案的原因，在于其方法更具适应性，并且能够利用更多类型的数据。银行和其他金融机构必须检测、预防并报告未经授权和非法金融活动，使用 AI/ML 处理这一问题的金融机构数量也在增加。银行和其他金融机构

可以用其生成的大量消费者行为数据训练模型，使 ML 模型学习欺诈模式，并在实践中检测欺诈行为。同样，在反洗钱价值链中使用该技术也可能带来收益，包括在客户准入、客户筛查以及交易监测中获得特别且直接的好处。此外，研究者认为，该技术还有助于减少误报，使银行能够把资源投入到真实欺诈事件中。

自然语言处理是 AI 的一个分支，它训练计算机像人类一样理解和处理语言。NLP 可用于开发大型语言模型，使其根据提示生成文章或计算机代码，其方法是基于某些概率预测词语序列中的下一个词。这项技术支持金融中的多种应用，包括驱动聊天机器人、检索企业自身的大量数据，以及编写代码。大型语言模型是人工神经网络的例子，它们试图通过把神经元复制为数学表征，并把这些人工神经元组织为层或簇，来反映人脑的运行方式。除了金融专用应用之外，大型语言模型，尤其是 ChatGPT，也让普通公众首次获得了具体且可直接使用的 AI 示例，而这些能力此前通常只为专业人员所用。